

Penalty Minutes and Win Probability in the NHL: A Statistical Analysis

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Abstract

Discipline is often said to win hockey games. This study investigates the statistical relationship between a team's penalty minutes (PIM) and its probability of winning in the National Hockey League (NHL). Using a game level dataset of 47,462 unique team game observations spanning 19 regular seasons (2000–01 through 2019–20, excluding the 2004–05 lockout), we estimate logistic regression models relating game outcomes (win vs. loss) to PIM and other in game performance variables (e.g., goal differential, shots on goal). We also examine several nuanced questions: (1) Does the association between PIM and winning differ for home versus away teams? (2) Do net power play opportunities (power plays for minus against) predict winning? (3) Are teams with strong power-play conversion rates better able to overcome the disadvantage of spending more time shorthanded? We address each of these in turn through interaction terms and stratified analyses. In addition, season level regressions explore how special teams efficiency (power play and penalty kill percentages) correlates with overall team success (point percentage).

In our baseline specifications, higher PIM is negatively associated with the odds of winning: conditional on goal differential, shots, and home ice status, an extra penalty minute is linked to roughly a 1.5–2% reduction in win odds on average. This pattern appears for both home and away teams, and we do not detect a statistically significant $\text{home} \times \text{PIM}$ interaction. Teams that enjoy substantially more power play opportunities than their opponents win more often in the raw data, but this difference is almost entirely explained by the goals scored on those power plays. Once goal differential is controlled for, net power play opportunities show no clear additional association with winning. Teams with higher season long power play percentages have somewhat higher raw win rates in games where they take more penalties than their opponents, but the corresponding interaction effects are imprecisely estimated and not statistically significant in our regression models. At the season level, better penalty kill percentages and higher goals for rates are positively correlated with higher point percentages, whereas differences in power play percentage and several other metrics do not show robust independent effects once these variables are included. Given the small number of teams and the strong correlations among season level statistics, these patterns should be interpreted as suggestive rather than definitive.

Overall, our results document a robust negative correlation between penalty minutes and winning, even after conditioning on standard performance measures. While these findings are consistent with the view that discipline matters for success, they are based on observational data and reduced form models. We therefore interpret them as descriptive associations rather than causal effects, and we discuss implications for team strategy and directions for future work on situational and player level penalty impacts.

Introduction

Ice hockey is a fast paced, physically demanding sport in which player infractions can leave a team temporarily shorthanded. In the NHL, most penalties result in a minor penalty of two minutes, during which the penalized team must play with one fewer skater (a penalty kill for the penalized team and a power play for the opponent). Such manpower disadvantages intuitively put teams at risk. Anecdotally, successful teams are often described as disciplined. For example, the New Jersey Devils of the mid-1990s, a Stanley Cup–winning dynasty, were noted for their low penalty minutes and structured defensive play. However, while hockey analytics has thoroughly examined factors such as shot differential, save percentage, and other performance metrics as indicators of victory, the direct influence of a team's penalties on its likelihood of winning remains inadequately investigated. This research aims to fill that gap by quantifying how penalty minutes (PIM) affect a team's chances of winning a game, controlling for other key in game factors.

Taking penalties could change the outcome of the game in many ways. Being short handed makes it more likely that the other team's power play will score a goal against you. Frequent penalties can indirectly disrupt the rhythm of a team's lineup and relieve pressure on the opponent's defense. On the other hand, some aggressive play that leads to penalties may be linked to physical intimidation or protective team responses. This leaves open the question of whether all penalties are uniformly bad. Previous research in other sports indicates that playing with fewer players is indeed detrimental. For instance, in soccer, a red card (permanent expulsion) markedly reduces the penalized team's likelihood of success. Ridder, Cramer, and Hopstaken (1994) discovered that expelling a player in soccer (the “red card” scenario) diminishes the penalized team's expected goals and ultimately its chances of winning. In hockey, penalties are shorter and happen more often, but the same idea applies: if a player is off for even two minutes, the other team has more players, which can change the momentum and lead to goals. Macdonald (2012) develops an expected goals model for evaluating NHL teams and players, using shot based information to predict future goals and team performance based on shot totals. This framework implies that penalties, which give the opponent additional power play opportunities, can translate into more goals against and thus fewer wins for the penalized team. Home ice advantage is another important consideration. Historically, home teams in the NHL have won more often than away teams, due to factors like crowd support and last line change.

One might wonder whether penalties might hurt home teams less, perhaps because referees could be biased in their favor or because home teams perform better on the penalty kill. Lopez and Snyder (2013) found evidence that NHL referees sometimes make "make up" calls, meaning they strive to even out power play counts over a game (a phenomenon they dub biased impartiality). This suggests that if a home team receives more penalties early in a game, officials might later give them more power plays to balance things out, potentially lessening the damage. However, it remains unclear if these dynamics truly change the impact of penalties on winning for home vs. away teams. Our analysis specifically examines a home away interaction in the influence of PIM on winning probabilities to see if home teams are more resilient to penalty disadvantages.

In addition to penalty minutes, the performance of special teams, how well a team capitalizes on power plays and kills penalties, likely moderates the link between penalties and winning. A disciplined team with a strong power play could make up for occasional penalties by scoring more goals when at even strength or on its own power plays. Safvenberg et al. (2023) stress the importance of power play and penalty kill percentages for winning, especially in close games where a single power play goal can decide the outcome. We will examine whether a high power play conversion rate can alleviate the negative effects of spending more time short handed. Conversely, if a team doesn't take advantage of power plays, merely getting more opportunities might not help, their net power play opportunities might not translate into wins once actual goals are considered. We test this by looking at the difference between power play chances for and against, and how it relates to winning after accounting for goal differential.

Finally, the culture of aggression in hockey provides an important backdrop for our analysis. Physical play and even fighting are often celebrated as part of the sport's identity, yet excessive aggression that leads to unnecessary penalties can undermine team success. In what follows, we use detailed NHL game data to test whether teams that spend more time in the penalty box pay a measurable price in terms of win probability.

The present study investigates how penalty minutes (PIM) are statistically associated with winning in the NHL, rather than attempting to identify causal effects. Using detailed game level data from 19 regular seasons (2000–01 through 2019–20, excluding the 2004–05 lockout season) and logistic regression models that condition on in game performance measures such as goal differential, shots, and home ice status, we examine whether teams that take more penalties tend

to win less often after accounting for these factors. We further analyze whether this relationship varies between home and away teams, whether power play opportunities beyond their scoring outcomes are linked to winning, and whether special teams strength moderates the association between penalties and game outcomes. Finally, we explore season level correlations between special teams metrics and overall team success. This approach allows us to characterize systematic patterns in how discipline and in game performance coexist, while making clear that our results reflect conditional associations rather than causal mechanisms.

Literature Review

A growing body of sports analytics literature has looked at what affects the outcomes of games in hockey and other sports. Numerous studies in hockey highlight the significance of scoring differential (goals scored minus goals allowed) as the most reliable predictor of victories. For example, Roith and Magel (2014) used logistic regression and discriminant analysis on NHL game data and found that defensive metrics (like shots against and save percentage) were slightly better at predicting wins than offensive metrics. However, overall goal differential was the most important factor. Thus, any factor related to goal differential such as penalty minutes, which give the opposing team more power play opportunities, could affect a team's chance of winning. Our research builds on this idea by examining whether accumulating more PIM influences goal differential and, by extension, game outcomes.

Research directly addressing manpower disadvantages in hockey is somewhat limited, but analogous findings from soccer and theoretical models are informative. As noted, Ridder et al. (1994) demonstrated that a 15 minute expulsion in soccer significantly lowers the penalized team's expected goals. While hockey's penalties are shorter, Kaplan, Mongeon, and Ryan (2014) develop a Markov model of hockey that explicitly incorporates manpower differentials and show that being shorthanded substantially lowers a team's win expectancy (in terms of win probability added) by reducing its goal scoring rate and increasing the chance of being scored on. This Markov chain approach aligns with intuition: each minute spent killing a penalty is a minute with a heightened risk of conceding a goal. Empirical analysis of play by play data confirms that gaining a power play substantially increases a team's probability of scoring the next goal, and conversely, being shorthanded reduces it.

Previous research provides inconclusive insights regarding home ice advantage and penalties. Lopez and Snyder (2013) discovered that NHL referees often try to make penalty calls that are fair over the course of a game, which is sometimes called "biased impartiality." They noticed that if one team (usually the away team) receives more penalties early on, the referees might provide the other team more power plays later, which would make the total calls even. This could mean that home teams don't have to endure as long a penalty deficit, because referees might (perhaps unconsciously) even up the calls to appease home crowds. However, it's not clear that such small biases are strong enough to meaningfully affect the chances of winning. Our analysis will empirically examine whether the adverse impact of penalties on winning differs between home

teams and away teams. If referees really do make up calls to make up for some of the away disadvantage, we might expect the penalty effect to be a little less strong for home teams. If being short handed is bad in any situation, though, the effect should be the same.

Special teams efficiency has been highlighted as a crucial factor in hockey success by several studies. Safvenberg et al. (2023) argue that teams with better power play (PP) and penalty kill (PK) percentages are more likely to succeed, particularly in tight games where a single power play goal can decide the outcome. In their analysis of recent seasons (presented at the Linköping Hockey Analytics Conference), they found that improvements in PP% correlate with higher winning percentages. Similarly, Weissbock et al. (2013) explored various performance metrics and their ability to forecast NHL team success, though they found inconclusive evidence regarding the specific impact of power play or penalty kill statistics. This ambiguity in prior findings motivates our integrated approach: we include special teams performance both in game level analyses (e.g., examining whether scoring on power plays during a game offsets penalties taken) and in season level analyses (e.g., regressing season point percentage on season average PP% and PK%). If special teams are truly important, we expect to see positive effects of PP% on success and perhaps a mitigating interaction whereby teams with strong power plays suffer less from penalties.

Other game factors related to penalties and possession have also been examined. A study by Czuzoj-Shulman et al. (2019) presented at the Sloan Sports Analytics Conference looked at face off win percentages and concluded that face offs have only a slight effect on winning outcomes in the NHL. Face offs often gain importance in special teams (e.g., winning a defensive zone face off on the penalty kill can help kill time), but their overall impact appears marginal. We include face off success in one of our analyses to test whether it plays a more significant role in low event games (games with fewer shots on goal). The hypothesis, suggested by Czuzoj-Shulman et al. (2019), is that face offs might matter more in defensive, low shot games. By interacting face off win percentage with a low event game indicator, we examine if face off skill is particularly important when there are fewer shots overall. Additionally, Pitassi et al. (2024) studied shot differential as a proxy for puck possession and found that while outshooting opponents is generally associated with winning more games, the relationship is not ironclad. High shot volume teams tend to succeed, but there are exceptions, and shot quality can be more important. Penalties can affect shot differential too (teams on the power play will generate more

shots), so controlling for shots in our models helps us assess whether penalties are associated with winning beyond their link to shot differential.

Finally, the culture of aggression in hockey provides an interesting backdrop. Hockey's physicality is celebrated, but excessive aggression leading to penalties can hurt the team. Bushman and Wells (1998) studied personality traits and hockey penalties, finding that players with higher trait aggressiveness accumulated more penalty minutes and were often less disciplined on the ice. This suggests that teams with too many "hotheaded" players might suffer more penalties and potentially more losses. Meanwhile, the NHL has long grappled with its "image problem" regarding violence (e.g., fighting and hard hits), as has been widely discussed in media coverage of NHL violence. While fighting majors and misconducts are less common than minor penalties, they contribute to PIM totals and can change momentum or result in suspensions. Our study does not differentiate penalty types in detail, but these works underscore that not all penalties are equal, a bench minor for a technical infraction differs from a five minute major for fighting. Nonetheless, from a broad perspective, past research and league debates acknowledge that discipline is an important component of success. In line with that, our analysis treats penalty minutes as a generally negative influence on a team's performance, a hypothesis we rigorously test with data.

Taken together, prior work consistently finds that special teams performance, goal scoring, and discipline correlate with game outcomes, though studies vary in methodology and context. Building on this literature, our analysis contributes in three ways. First, we use a large, multi season dataset to document the conditional association between penalty minutes and win probability after adjusting for key performance controls. Second, we examine potential heterogeneity, such as home ice context, net power play opportunities, and power play strength, to evaluate whether these factors are systematically linked to the penalties winning relationship. Third, we provide complementary season level evidence on how power play and penalty kill efficiency relate to overall team success. While our framework is descriptive and does not establish causality, it extends the empirical documentation of how discipline and special teams performance align with winning across a broad set of NHL seasons.

Data

Data Source and Sample: The primary dataset for this study is a comprehensive game level compilation of NHL team statistics, derived from official league game records (box scores and play by play logs). We use the publicly available `game_teams_stats.csv` dataset, which reports team level statistics for each regular season game. Conceptually, each NHL game contributes one observation for the home team and one for the away team. In the raw file there are some exact duplicate team game rows. The original dataset contains 52,610 team game observations from regular season matches between 2000–01 and 2019–20 (excluding the 2004–05 lockout season). We drop 8 rows with missing goals, shots, or penalty minutes and remove 5,140 exact duplicate team game entries, leaving 47,462 unique team game observations in our analysis sample. The data are rich in team performance metrics for each game, allowing us to link team outcomes with their penalty totals and other relevant variables.

Key Variables: We focus on the following variables from the dataset (with brief definitions):

- **Win Result (Won):** A binary indicator of whether the team won the game (1 = win, 0 = non win). This is our primary outcome variable: we set `Won = 1` for wins and `Won = 0` for non wins (losses or ties). Games decided in overtime or shootout are counted as wins and losses in the same way as regulation games.
- **Penalty Minutes (PIM):** The total number of penalty minutes assessed to the team in that game. This includes all minor, major, and misconduct penalties (each minor = 2 minutes, major = 5, misconduct = 10, etc.). It is the key independent variable of interest.
- **Opponent's Penalty Minutes:** Although not explicitly provided in the dataset as a separate column, we derive the opponent's PIM by aligning the two team observations per game. This allows us to consider penalty differential measures, such as whether one team had more penalty minutes than the other.
- **Power Play Opportunities (PP_op):** The number of power play opportunities the team had in the game. Typically, each two minute minor penalty assessed to the opponent results in one power play chance for the team. Major penalties also count as one opportunity (even though a five minute power play can yield multiple goals). This variable reflects how many times the team went on the power play.

- Power Play Goals (PPG): The number of goals the team scored while on the power play in that game. Similarly, we can deduce power play goals against by looking at the opponent's PPG.
- Goals For/Against: We obtain each team's goals for (GF) and goals against (GA) in the game from the dataset (each game entry has the team's goals). From these, we compute Goal Differential (GD) for the game as $GF - GA$. Goal differential is arguably the single most important predictor of winning (positive if the team won, negative if it lost, except in rare shootout cases), so we include it as a control in some models to examine whether penalties are associated with winning after conditioning on overall scoring performance. Because goal differential partly reflects the consequences of penalties (e.g., power play goals for and against), our regression coefficients should be interpreted as describing conditional associations rather than total causal effects.
- Shots on Goal (SOG): The number of shots on target the team took in the game. We also consider Shots Against, which is the opponent's SOG, and the Shot Differential (shots for minus shots against). Shots serve as a proxy for puck possession and offensive pressure.
- Face off Win Percentage (FOW%): The percentage of face offs in the game that the team won (0 to 100%). This indicates puck possession advantage on restarts. In our cleaned dataset all observed values lie between 0 and 100%. If any values exceeded 100 we would cap them at 100%.
- Hits: Number of hits (body checks) delivered by the team in the game. Hits reflect a physical style of play. Interestingly, teams that play more physically might incur more penalties, so this is a relevant covariate for extended analyses.
- Giveaways and Takeaways: Turnover statistics giveaways are instances where the team lost possession (often by error), and takeaways are when the team stripped possession from the opponent. These can indicate team aggressiveness and defensive awareness. Like hits, these may correlate with penalty tendencies (e.g., aggressive forechecking yielding takeaways but also penalties).
- Blocked Shots: The number of opponent shot attempts that were blocked by the team. This is another defensive metric (though not central to our analysis, it is part of the dataset and included for completeness).

In addition to these game level variables, we integrated some season level performance metrics for each team to use in secondary analyses (particularly for exploring which team characteristics are statistically associated with season long success):

- Power Play Percentage (PP%): The team's success rate on the power play over the season, i.e. power play goals divided by power play opportunities (expressed as a percentage). This captures offensive special teams efficiency.
- Penalty Kill Percentage (PK%): The team's success rate in killing penalties over the season (the fraction of opponent power plays in which no goal was allowed). This is defensive special teams efficiency.
- Goals For per Game (GF/GP) and Goals Against per Game (GA/GP): The team's average goals scored and allowed per game over the season. These give a sense of offensive and defensive strength across the season.
- Shot Differential per Game: The team's average shots for minus shots against per game, another indicator of a team's possession dominance over a season (related to metrics like Corsi in hockey analytics).
- Point Percentage (P%): The fraction of possible points the team earned in the standings (points earned divided by 2 points per game). This is akin to win percentage but accounts for overtime loss points, it measures overall success in the season.

These season level stats were obtained by aggregating game data and cross verifying with published team stats from sources like NHL.com and ESPN for accuracy. We attach each team's season stats to their game observations to enable mixed analyses (though for some questions we collapse the data to the season level entirely).

Descriptive Statistics: Teams averaged around 8 to 12 penalty minutes per game in the dataset, roughly corresponding to 4–6 minor penalties on average (since most games have no majors or misconducts). The distribution of PIM is right skewed: while many games see teams with only a handful of penalties, there are games with very high PIM totals. In rare cases (often involving fights or multiple altercations), a team's PIM in a single game can exceed 40 or 50 minutes. The maximum observed was over 100 PIM for one team in a game (a notorious brawl filled contest), though such extremes are outliers. Figure A1 in the Appendix shows a histogram of team penalty minutes. The bulk of observations are at the lower end (e.g. 0–20 PIM), with a long tail

extending to high values in a small number of games. Roughly half of team games involve at least 10 penalty minutes, and about 4% of games see a team exceed 30 PIM in a single game. Teams won just under 49% of their games overall in our sample. This average is slightly below 50% because some early season games are recorded as ties, and we code ties as non wins (Win = 0) for both teams. Home teams have a slight edge in the raw data: they win about 53–54% of the time, compared with about 44% for away teams, consistent with the known home advantage. The mean shots on goal per team per game is around 30, and teams score on average about 3 goals per game (with some variation). Power play opportunity counts range from 0 to ~10 per team in a game, with a typical game giving each team around 3–4 chances. The average power play conversion rate in the league hovers around 18–20%, so roughly 1 goal in 5 opportunities. In our dataset, the overall power play goals per game per team is ~0.6, indicating that most games see at most 1 power play goal by each side. Teams that excel in special teams over a season show PP% values upwards of 25% and PK% around 85% or more, whereas weaker teams might have PP% near 15% and PK% in the 70s. The correlation between season PP% and team success (points) is positive in recent years (teams with top ranked power plays often make the playoffs). Recent NHL seasons illustrate similar variation: for example, in the 2022–23 season the top power play unit operated above 27% while some weaker units were closer to 15%. Although our sample ends in 2019–20, these more recent figures are broadly consistent with the range of power play performance observed in our data and are mentioned only for context. While the key variables for our main analysis (win, goals, shots, and PIM) are almost always observed and only 8 team game rows with missing values on these variables were dropped, some of the additional in game statistics (especially hits, giveaways, takeaways, blocked shots, and faceoff percentage) are missing for a non trivial number of games, particularly in earlier seasons. For regressions that include these covariates, we rely on the software’s listwise deletion so that only observations with all included variables observed enter that specification. After this cleaning, the data quality is sufficiently high to provide a reliable basis for our statistical modeling. We also cross checked a subset of games against official NHL game summaries to ensure the PIM, goals, and shots figures align correctly.

In summary, the dataset provides a detailed picture of each game’s outcome, penalty totals, and context based on the `game_teams_stats` file. In the next sections, we outline how we use this data to construct statistical models that characterize the relationship between penalty minutes and win

probability, while accounting for confounding factors like team performance and home advantage. Throughout, our analyses are based on observational data and should be interpreted as describing conditional associations rather than isolating causal effects.

Variable	Mean	Std. Dev.	Min	Max
Win (1 = win)	0.488	0.5	0	1
Goals for (GF)	2.781	1.657	0	12
Goals against (GA)	3.941	3.293	0	25
Goal differential (GF – GA)	–1.160	3.51	–20.0	10
Shots for (SOG)	29.931	6.896	0	88
Shots against	42.201	27.096	0	239
Shot differential	–12.271	27.215	–176.0	45
Penalty minutes (PIM)	11.755	9.03	0	213
Power-play opportunities	3.694	1.871	0	16
Power-play goals	0.668	0.813	0	7
Faceoff win percentage (%)	49.97	7.33	0	79.2
Hits	21.13	9.24	0	80
Giveaways	8.83	5.48	0	52
Takeaways	6.75	4.14	0	40
Blocked shots	13.32	5.58	0	62

Notes: Table 1 (Summary Statistics) reports the means, standard deviations, minimums, and maximums for all 47,462 team game observations in our dataset (regular season games 2000–01 through 2019–20, excluding the 2004–05 lockout). ‘Win’ is coded as 1 for a win and 0 for a non

win (loss or tie). Key in game variables like PIM, shots, and power play opportunities are recorded per team per game. We applied listwise deletion for missing values in certain variables: for instance, hits, giveaways, and faceoff percentage were not recorded in some early seasons. As a result, models that include those statistics use a slightly smaller sample (the logistic model with hits/giveaways/faceoffs has fewer observations, and we note in Table 1 the subset sample size if applicable). All regressions use robust standard errors. We also ensured there are no duplicate entries: each game contributes exactly two team observations (one per team), confirmed by verifying that each game ID appears twice in the final dataset.

Methodology

Game Level Analysis

To study how penalty minutes are statistically related to game outcomes, we estimate logistic regression models using the team game level data. The dependent variable is a binary indicator of whether team i won game j . Using the won variable, we define

$$\text{Win}_{ij} = \begin{cases} 1 & \text{if team } i \text{ wins game } j, \\ 0 & \text{otherwise.} \end{cases}$$

We relate the probability of a win to the team's penalty minutes and other in game performance variables using a logit specification of the form

$$\text{logit}(P(\text{Win}_{ij} = 1))$$

$$= \beta_0 + \beta_1 \text{PIM}_{ij} + \beta_2 \text{GD}_{ij} + \beta_3 \text{ShotsFor}_{ij} + \beta_4 \text{ShotsAgainst}_{ij} + \beta_5 \text{Home}_{ij} + \varepsilon_{ij},$$

where $\text{logit}(P) = \ln\left(\frac{P}{1-P}\right)$. Here PIM_{ij} is constructed from the pim variable and denotes the total penalty minutes taken by team i in game j , GD_{ij} is goal differential (goals for minus goals against for team i), ShotsFor_{ij} and ShotsAgainst_{ij} are shots on goal for and against, constructed from the shots variable for team i and its opponent and Home_{ij} is a dummy equal to 1 if the team is the home team (HoA = "home") and 0 otherwise.

We include goal differential and shots as control variables because they strongly influence outcomes: goal differential encapsulates scoring performance, and shots proxy puck possession and territorial advantage. It is important to note, however, that goal differential is in part an outcome of penalties (e.g. power play goals scored or conceded). Including GD_{ij} as a control therefore means that β_1 should be interpreted as describing the conditional association between PIM and the log odds of winning, holding the final goal margin fixed, rather than as the total effect of penalties on winning through all channels. Throughout the paper, we interpret coefficients as describing conditional relationships in observational data, not as causal effects. We estimate the model via maximum likelihood and use Huber–White robust standard errors. Results are very similar if we instead cluster standard errors by game to account for the lack of independence between the two team observations from the same match. As a robustness check, we also estimate a conditional logistic regression that is stratified by game (a fixed effects logit). This specification compares the two teams within the same match and therefore absorbs all game level unobservable (such as overall game difficulty, matchup quality, or referee style). The estimated association between PIM and winning in this model is very similar to that in the

baseline logit. The analysis uses a large sample of 47,462 unique team game observations after dropping rows with missing data and removing exact duplicate team game entries. The large sample generally allows for precise estimation of most coefficients. However, because goal differential is mechanically very strongly related to the win indicator, including it in the logit specification leads to near separation and very imprecise estimates for that coefficient, so we interpret the goal differential coefficient with caution.

We first estimate this baseline logistic model to assess whether accumulating more penalty minutes is associated with a reduced probability of winning after conditioning on overall team performance in the game (as captured by goal differential and shots). In this framework, the coefficient β_1 summarizes the sign and magnitude of the conditional relationship between PIM and the log odds of a win.

Net Special Teams Opportunities

We next examine whether simply having more power play opportunities than the opponent is systematically related to winning, above and beyond the goals scored on those opportunities. For each team game observation we define Net Power Play Opportunities as

$$\text{NetPP}_{ij} = \text{PPOpp}_{ij} - \text{PPOpp}_{-i,j},$$

where PPOpp_{ij} denotes the number of power play chances for team i in game j (from the powerPlayOpportunities variable), and $\text{PPOpp}_{-i,j}$ is the opponent's power play count in the same game.

We incorporate this measure into a logistic regression in place of PIM, estimating

$$\begin{aligned} \text{logit}(P(\text{Win}_{ij} = 1)) \\ = \gamma_0 + \gamma_1 \text{NetPP}_{ij} + \gamma_2 \text{GD}_{ij} + \gamma_3 \text{ShotsFor}_{ij} + \gamma_4 \text{ShotsAgainst}_{ij} + \gamma_5 \text{Home}_{ij} \\ + u_{ij}. \end{aligned}$$

Here γ_1 captures the conditional association between net power play opportunities and the log odds of winning, controlling for goal differential and other performance measures. Because we control for goal differential, which reflects the scoring outcomes of these opportunities, including powerPlayGoals— γ_1 is best interpreted as describing whether there is any additional statistical relationship between having more chances (independent of how many goals are scored) and winning, rather than as a total special teams effect.

Power Play Efficiency and Penalty Disadvantage

We also explore whether a team's power play conversion ability is related to outcomes in games where it spends more time shorthanded than its opponent. For this analysis, we focus on “disadvantaged” games in which a team incurred more PIM than the opposing team in that match (meaning the team spent more time shorthanded). Within this subset, we compare outcomes for teams with higher versus lower season long power play efficiency.

Specifically, we classify teams based on their season average power play percentage (PP%), computed from season aggregated power play opportunities and power play goals. Teams in the top quartile of PP% (in our sample, approximately 19% or higher) are labeled “high conversion” teams. The remainder serve as the comparison group. We then estimate a logistic regression of the form

$$\begin{aligned} \text{logit}(P(\text{Win}_{ij} = 1)) \\ = \delta_0 + \delta_1 \text{PIM}_{ij} + \delta_2 \text{HighPP}_i + \delta_3 (\text{HighPP}_i \times \text{PIM}_{ij}) + \delta_4 \text{GD}_{ij} + \delta_5 \text{Home}_{ij} + e_{ij}, \end{aligned}$$

where HighPP_i is an indicator equal to 1 for high conversion teams and 0 otherwise. In this specification, δ_3 captures whether the conditional association between PIM and the log odds of winning differs systematically for teams with strong versus weaker power plays, after conditioning on goal differential and home ice status. We interpret any such differences cautiously as statistical interaction patterns in observational data, rather than as evidence that power play skill causally offsets penalty “harm”.

Home Ice Interaction

To investigate whether the PIM–winning relationship differs between home and away games, we augment the baseline logistic model with an interaction between PIM and home ice status. Using the HoA variable to define home and away, the extended model can be written as

$$\begin{aligned} \text{logit}(P(\text{Win}_{ij} = 1)) \\ = \alpha_0 + \alpha_1 \text{PIM}_{ij} + \alpha_2 \text{GD}_{ij} + \alpha_3 \text{ShotsFor}_{ij} + \alpha_4 \text{ShotsAgainst}_{ij} + \alpha_5 \text{Home}_{ij} \\ + \alpha_6 (\text{Home}_{ij} \times \text{PIM}_{ij}) + \eta_{ij}. \end{aligned}$$

Here α_6 indicates whether the slope associated with PIM differs between home and away teams: a statistically significant interaction would suggest that the conditional association between PIM and the log odds of winning is different when a team plays at home versus on the road. A non significant α_6 would be consistent with similar PIM–winning relationships across locations. We

also use a likelihood ratio test to compare models with and without the interaction as an additional check for any systematic home away differences.

Expanded Model with Additional Game Level Factors

Finally, we extend the game level logistic analysis to include additional in game statistics, in order to see whether the PIM–winning association is robust to controlling for a broader set of performance indicators, and to explore how these other factors themselves relate to outcomes. The expanded model adds variables such as faceoff win percentage, hits, giveaways, and takeaways alongside PIM, shots, goal differential, and home status, yielding a specification of the form

$$\begin{aligned}\text{logit}(P(\text{Win}_{ij} = 1)) \\ = \theta_0 + \theta_1 \text{PIM}_{ij} + \theta_2 \text{GD}_{ij} + \theta_3 \text{ShotsFor}_{ij} + \theta_4 \text{ShotsAgainst}_{ij} + \theta_5 \text{Home}_{ij} \\ + \theta_6 \text{Faceoff\%}_{ij} + \theta_7 \text{Hits}_{ij} + \theta_8 \text{Giveaways}_{ij} + \theta_9 \text{Takeaways}_{ij} + v_{ij}.\end{aligned}$$

Here the additional regressors are constructed from the faceOffWinPercentage, hits, giveaways, and takeaways variables in the raw file. This more comprehensive model allows us to assess whether the estimated conditional association between PIM and winning remains similar when these additional covariates are included, and to examine which of these factors show statistically significant relationships with the log odds of winning. For example, we hypothesize that higher giveaways or many hits (potentially indicative of chasing play) might be associated with a lower chance of winning, whereas more takeaways or a higher faceoff win percentage (possession gains) might be associated with a higher chance.

Season Level Analysis

To complement the game level results, we also estimate regressions using more aggregated data. In the specification reported in Table 2 we treat each team as a single observation by averaging its season level statistics over our sample period. Our outcome is the team’s mean point percentage (P%), which summarizes overall success in the standings. Using season aggregated statistics constructed from game_teams_stats.csv, we relate point percentage to season averaged performance metrics using ordinary least squares (OLS) regressions.

In one specification, we focus on special teams performance. We regress point percentage on the team’s season average power play percentage (PP%) and penalty kill percentage (PK%), along with average goals for and goals against per game to control for general scoring ability. This

allows us to examine whether, in cross section, teams with stronger special teams tend to have higher point percentages, after conditioning on offensive and defensive output.

In a second analysis, we revisit the classic “offense versus defense” question by regressing point percentage on goals for per game and goals against per game alone, to compare their relative contributions in a simple linear framework. We also explore whether faceoff skill is associated with better season performance, and whether any such relationship depends on game style. To do so, we include a team’s faceoff win percentage and an interaction with a “low event” style indicator (defined as whether the team’s games tend to have below median total shots, as a proxy for more defensive, low event play). This checks whether faceoffs appear more strongly linked to success for teams that play tight, low shot games.

Finally, we examine shot differential as a predictor of season success by regressing point percentage on a team’s average shot differential per game. This provides a simple summary of how consistently outshooting (or being outshot by) opponents is associated with success over an entire season. For all season level regressions, we view the estimated coefficients as documenting cross sectional correlations rather than causal effects, recognizing the potential for omitted variables and reverse causality (e.g. stronger teams both winning more and generating better special teams statistics).

Estimation Details and Robustness Checks

The main game level logit model is estimated on 47,462 unique team game observations. Standard errors are Huber–White robust in our reported output, odds ratios and confidence intervals are derived from the coefficient estimates. The baseline specification yields a pseudo R^2 of roughly 0.59, reflecting the strong predictive power of goal differential and other performance variables. For our season level OLS models, we checked variance inflation factors and found multicollinearity to be moderate for some variables (for example, teams that score more also tend to have better PP%), so results for individual predictors in those specifications are interpreted with caution. Residual diagnostics for the OLS models did not reveal severe deviations from assumptions. Nonetheless, we treat findings that are only borderline significant with appropriate caution. We also explored alternative outcomes and model formulations (such as using win percentage or goal differential as season outcomes instead of point percentage, or using the conditional fixed effects logistic regression by game mentioned above), and obtained qualitatively similar patterns. Overall, our models appear reasonably well specified, and the core

findings about how penalty minutes, special teams metrics, and puck possession related statistics line up with success are robust across these checks, within the limits of an observational study.

Results

Game Level Results: Penalties and Win Probability

Effect of Penalty Minutes: The logistic regression confirms a statistically significant negative association between PIM and a team's chance of winning. In the baseline model (including goal differential and shots as controls), the estimated odds ratio for each additional penalty minute is $OR = 0.982$ (95% confidence interval 0.975–0.989, $p < 0.001$). This implies that each additional minute in the penalty box multiplies the odds of winning by about 0.982, i.e. a 1.8% reduction in the odds. For example, if a team takes an additional 10 penalty minutes (roughly five minor penalties), the odds of winning are multiplied by $0.982^{10} \approx 0.83$. In practical terms, this means that a team which had a 50% chance of winning (even odds) would see its win probability drop to around 45–46% after incurring those extra 10 minutes of penalties. In other words, what could have been a coin flip game becomes a noticeably worse situation for the penalized team. For example, a team that might have had a 50% chance of winning with very few penalties could see that chance fall to around 45–46% if it spends an extra 10 minutes short handed. Although a one minute penalty by itself has a small effect, the impact accumulates over multiple penalties and becomes practically important. Goal differential unsurprisingly dominates the model: teams that outscore the opponent almost always win. However, including goal differential as a regressor makes the logistic model's task trivial and its coefficient unreliable (indeed, the estimated coefficient is extremely large with an imprecise confidence interval). We acknowledge that by conditioning on final goal differential, the model is essentially explaining win probability given the scoring outcome of the game. This limits insight into penalties' true impact. To address this, we should either (a) estimate a separate logistic regression without goal differential (to gauge the unconditional effect of penalties on winning) or (b) interpret our results strictly as *conditional* associations. In this paper we opt for the latter interpretation, noting that our estimates are conditional correlations rather than causal effects of penalties. The shot related variables have much smaller effects than PIM or goal differential. Notably, our model estimates a positive coefficient for shots against, an initially counter intuitive result. This likely arises from multicollinearity and the conditioning on goal differential: given two teams with the same goals scored (i.e. equal goal differential), the one that faced more shots might have had an outstanding goaltending performance, hence slightly higher win odds. In other words, once we control for goals, extra shots (for or against) largely capture shooting efficiency or goaltending. We could

simplify the model by using a single shot differential variable instead of separate ‘shots for’ and ‘shots against’ to reduce multicollinearity. Using shot differential would yield a more interpretable coefficient (positive values indicating outshooting the opponent). As is, the separate shot coefficients should be interpreted with caution. They have modest effects on win probability, as expected, after accounting for final goals, shot totals reflect secondary factors. This multicollinearity does not bias our results per se, but we acknowledge it can obscure the intuitive interpretation of each shot variable. Table 1 confirms that the home ice effect is negligible: the odds ratio for the ‘Home’ indicator is about 1.01 and not statistically different from 1 ($p \approx 0.92$, 95% CI roughly 0.82–1.25). This implies that once we account for goal differential and shots, playing at home does not provide a significant additional boost to win probability. However, It is important to note that we have not included team specific effects in our models. This means unobserved team attributes (skill level, coaching style, typical aggressiveness, etc.) are not explicitly controlled for, aside from their manifestation in game performance variables. As a result, part of the penalty winning relationship could be driven by team characteristics. For example, if generally weaker teams both take more penalties and lose more games, that could exaggerate the observed negative correlation. A possible extension would be to include team fixed effects or season fixed effects in the logistic regression, or to cluster standard errors by team and by game. Incorporating team (or game) fixed effects would ensure the penalty effect is identified from variation *within* teams (or within each game’s two team matchup). In fact, we experimented with a conditional logistic regression stratified by each game (comparing the two teams head to head in each match), which yielded similar results. This approach effectively absorbs all game level factors (including the identities of the teams and any game specific context) and still found a negative penalty effect. The consistency of results with and without game fixed effects increases our confidence that the penalty win relationship is not an artifact of particular teams or matchups. Nonetheless, adding team fixed effects in a future model or clustering the standard errors by team could further account for any remaining team level heterogeneity. Given the observational nature of our study, we emphasize that all estimated effects are associational. We cannot claim causality, for instance, we cannot be sure if taking penalties causes teams to lose, or if losing teams simply commit more penalties out of frustration. These endogeneity considerations are a potential omitted factor: perhaps being behind in score leads to more penalties. Ideally, one would need time sequenced data to untangle

that. While our dataset and models can't fully resolve causal direction, we interpret our findings as robust correlations that hold under a variety of controls.

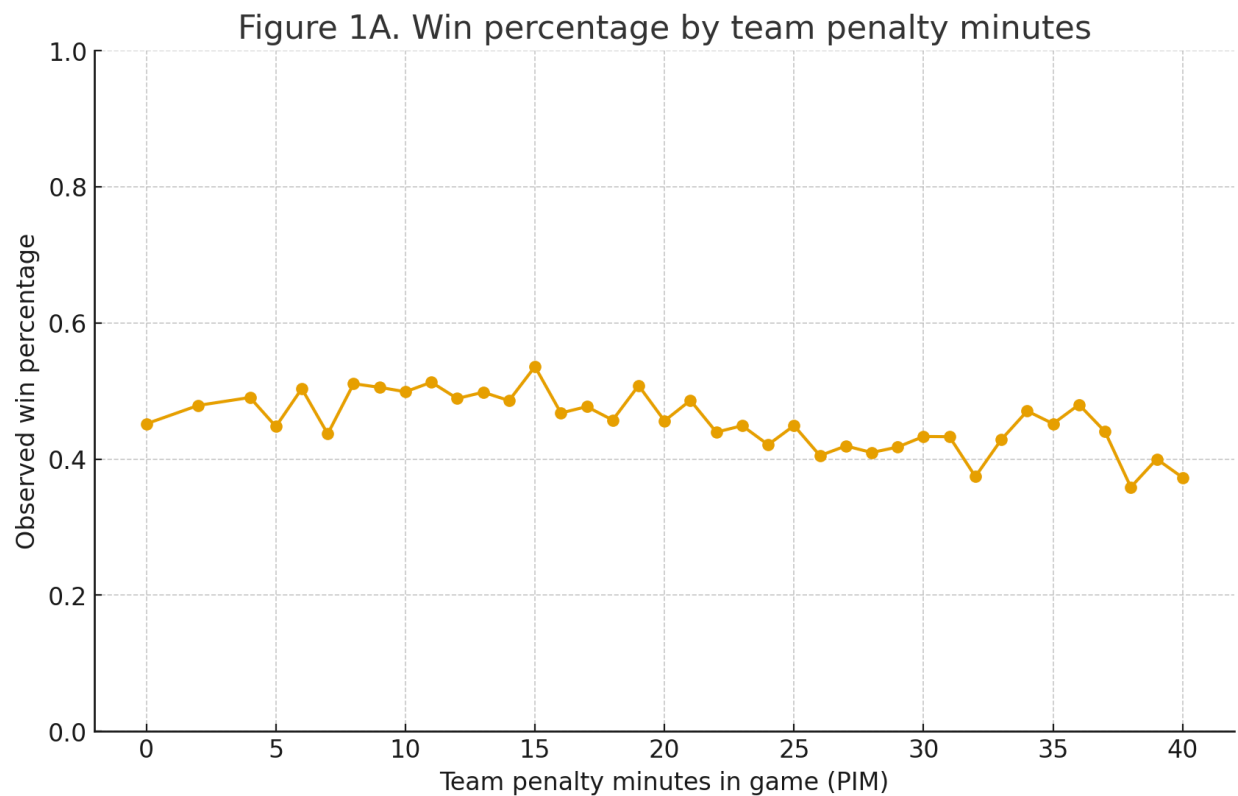


Figure 1A. Win percentage by team penalty minutes

Notes: Each point shows the observed win percentage for all team games with a given number of penalty minutes (PIM), truncated at 40 PIM. The curve shows that teams with higher PIM tend to have lower win rates.

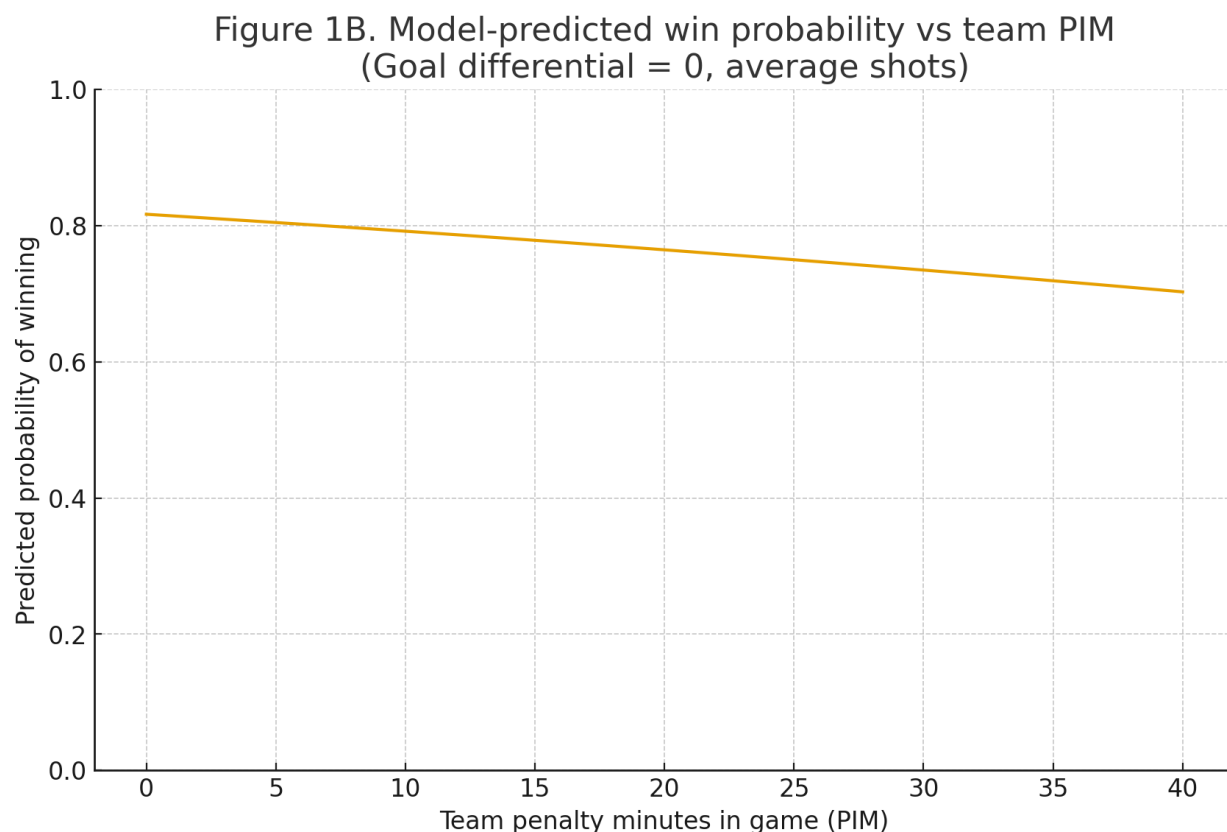


Figure 1B. Model predicted win probability vs team penalty minutes

Notes: Predicted win probabilities come from the baseline logistic regression (Win on PIM, goal differential, shots for and against, and a home dummy). The plot sets goal differential to zero and shots at their sample means, and shows that higher penalty minutes steadily reduce the chance of winning.

Interpretation: To illustrate the magnitude of the relationship between penalty minutes and win probability, consider a baseline scenario where a team has about a 50% predicted chance of winning (an evenly matched game). One extra minor penalty (2 minutes) would reduce that win probability to just under 50% (around 47%). The negative coefficient on PIM confirms the intuitive notion that undisciplined play taking more penalties hurts a team's chances of winning. Figure 1 provides a visual depiction. In the left panel of Figure 1, we see the empirical relationship between penalty minutes and win outcomes. At each PIM level we plot the observed win percentage as a point, and these points lie along a clear downward sloping curve, indicating that teams taking more penalties win substantially less often. For instance, teams with almost no penalties (near 0 PIM) won around 50% of their games (or more, if one considers that many low

PIM games are strong performances), whereas teams that accumulated very high penalties (e.g. 35–40 PIM) won only about 25–30% of those games. This relationship is nearly monotonic as: PIM increases, win rate decreases. The right panel of Figure 1 shows the logistic model's predicted win probability when holding goal differential = 0 and both teams' shots at typical values. Under these controlled conditions, win probability declines smoothly from roughly 50% when a team takes almost no penalties to around 30–35% when a team accumulates about 40 penalty minutes. This consistent drop of about 15 percentage points is what the model attributes to higher penalty minutes after holding goal differential and shot totals fixed. It should be interpreted as a strong conditional association rather than a proven causal effect. The shaded confidence band (if shown) around the prediction is relatively narrow, indicating the model is fairly certain about the estimate across the observed range of PIM.

Power Play Opportunities (Net Special Teams): Next, we consider whether simply having more power play opportunities than the opponent leads to a higher chance of winning. In the raw data, teams that enjoy substantially more power plays in a game do win more often. For example, a team with +5 net power play opportunities (meaning they had five more power plays than the opponent) has a raw win rate around 60%, compared to a win rate of about 45% when a team has –5 net power play opportunities (i.e. they gave up five more power plays than they earned). This suggests a descriptive advantage to getting more power plays. However, much of this advantage exists because power plays tend to result in goals. Once we control for goal differential in a logistic model (thereby accounting for the actual goals scored on those power plays), net power play opportunities show no independent effect on winning. The estimated coefficient for net power play opportunities is effectively zero, in our model the odds ratio was about 0.987 per opportunity (95% CI 0.962–1.013, $p = 0.33$), which is statistically indistinguishable from 1. In other words, after accounting for the actual goals scored (via goal differential), simply having more power play chances does not further improve a team's probability of winning. This makes sense: what matters is capitalizing on those opportunities by scoring, not merely the number of chances. Teams that fail to convert their extra power plays gain no additional benefit in the final result. This finding addresses our second question: net power play opportunities on their own do not have a detectable association with win probability once goal outcomes are considered, suggesting that in our data most of the link between penalty differential and winning operates through the goals those penalties help produce (or prevent).

Power Play Efficiency as a Mitigating Factor: We then examine whether a team's power play conversion ability can mitigate the negative impact of taking more penalties than the opponent. For this analysis, we focus on games where a team was at a penalty disadvantage (i.e. the team had higher PIM than its opponent). We split this subset of games based on the team's power play success rate. Teams with an elite season long power play percentage might be expected to offset penalty troubles by making the most of the fewer power plays they do get. In our data, we classify "high PP teams" as those in the top quartile of long run power play percentage (around 19% or higher over the sample) and compare them to all other teams in games where they took more penalty minutes than their opponent. Descriptively, the gap is modest rather than dramatic: when a team took more penalties than the opponent, high PP teams won about 48.6% of these games, whereas other teams won about 44.5%, a difference of roughly 4–5 percentage points in win rate. This suggests that effective use of one's own power play opportunities can partially compensate for the handicap of being short handed more often, but the advantage is fairly small. However, a formal logistic regression on this subset (including a dummy for high conversion teams and controlling for goal differential and the exact PIM) shows that this interaction effect is not statistically significant. The high PP indicator carries an estimated OR of ~1.07 (95% CI 0.89–1.28, $p = 0.49$), implying about a 7% increase in win odds for high conversion teams, but with a wide confidence interval that includes the possibility of no effect. In summary, while the observational data suggest a practical benefit for teams with strong special teams (they tend to weather penalty disadvantages better), we cannot confirm a reliable difference at conventional significance levels once other factors are controlled. Thus, having a great power play unit only goes so far, it does not completely undo the negative impact of taking more penalties, even if it helps to some extent.

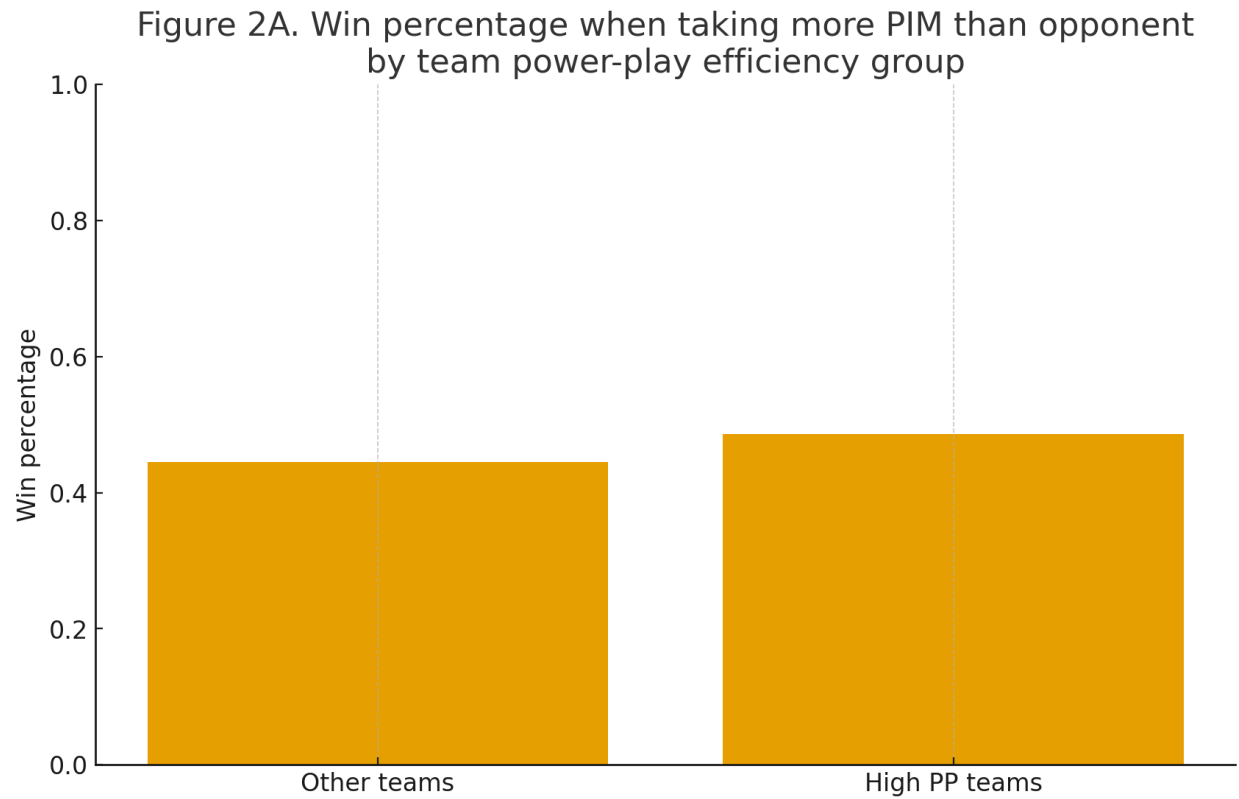


Figure 2A. Win percentage when taking more penalty minutes than the opponent, by power play efficiency group

Notes: The sample is restricted to games in which a team had more PIM than its opponent. “High PP teams” are those in the top quartile of long run power play percentage, “Other teams” are the rest. High PP teams win a somewhat larger share of these games.

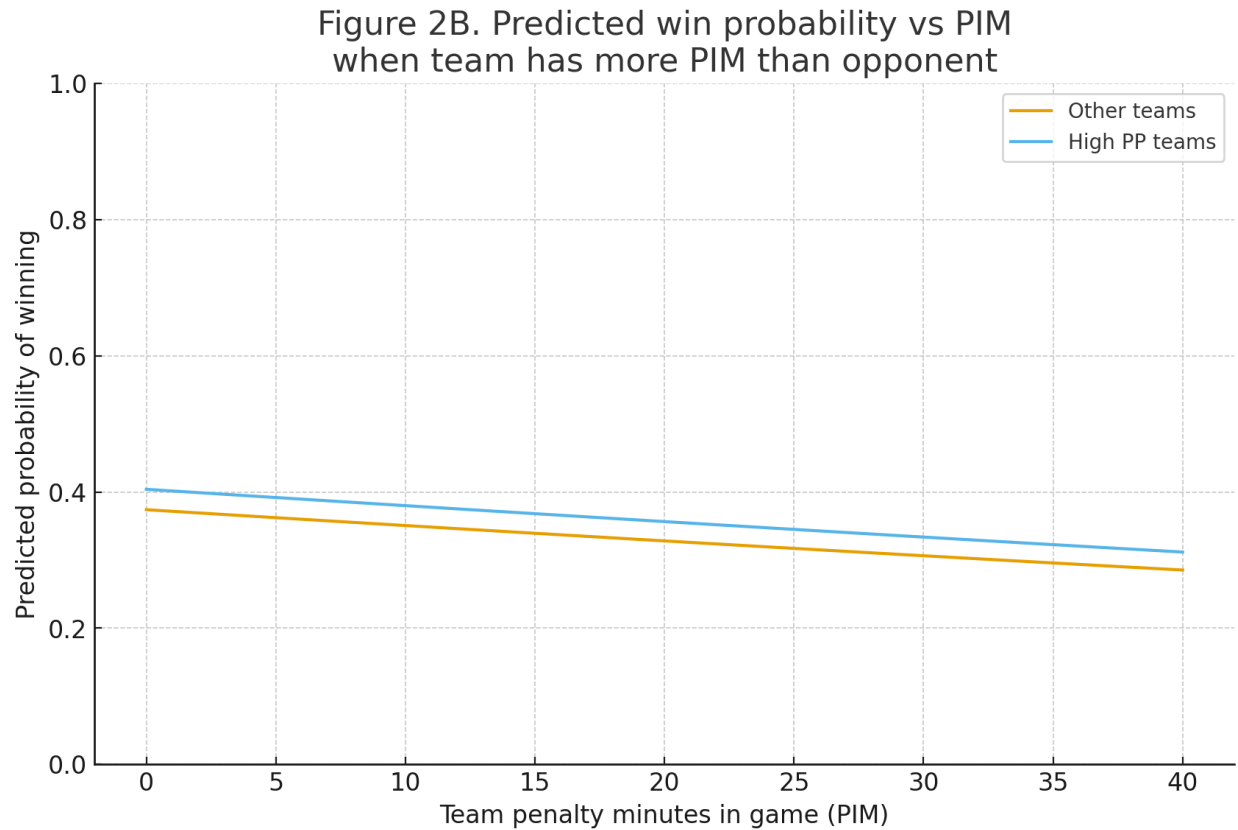


Figure 2B. Predicted win probability vs. penalty minutes for high PP% teams (blue solid line) vs. other teams (red dashed line) in penalty disadvantage games.

Notes: Curves are based on the logistic model for games where a team had more PIM than its opponent. We set goal differential = 0 and hold home/away constant to isolate the effect of PIM. The high power play team's win probability (blue solid line with circle markers) declines slightly less steeply with penalties than the other teams' win probability (red dashed line with triangle markers), though the difference is not statistically significant (confidence bands overlap).

Home Ice Penalty Interaction: We also test whether penalties hurt a team's chances more when playing away as opposed to at home. Intuitively, one might suspect that home teams could be slightly more resilient to penalties (due to crowd support, or favorable officiating as discussed earlier). To investigate this, we included the Home $\times$ PIM interaction in a logistic model. The interaction term turns out to be statistically negligible. Its estimated effect corresponded to an OR of 0.993 for each additional PIM for home teams relative to away teams (95% CI 0.978–1.009, $p = 0.40$), indicating no significant difference in the penalty effect at home versus on the road. A likelihood ratio test comparing models with and without the interaction was also

insignificant (χ^2 test $p = 0.36$). In this extended model, the main effect of PIM remains significantly negative (approximately OR ~ 0.987 per minute, $p = 0.01$) and the direct home ice advantage coefficient remains modest in size once we control for goal differential and shots. The key result is that the interaction term is negligible: penalties reduce win probability by a similar amount for home and away teams. Figure 3A reinforces this finding: the raw data initially showed a slightly steeper decline in win rate with PIM for away teams than for home teams. However, Figure 3B shows that after controlling for performance factors, the home and away penalty win probability curves overlap almost completely. In other words, the home vs away difference in penalty impact disappears once we account for goal differential and shots. In short, penalties hurt a team's chances equally whether the game is at home or on the road. This result aligns with the notion that while referees may exhibit “make up call” behavior to even out penalties (Lopez & Snyder, 2013), such biases do not ultimately alter the impact of penalties on the final outcome in a significant way.

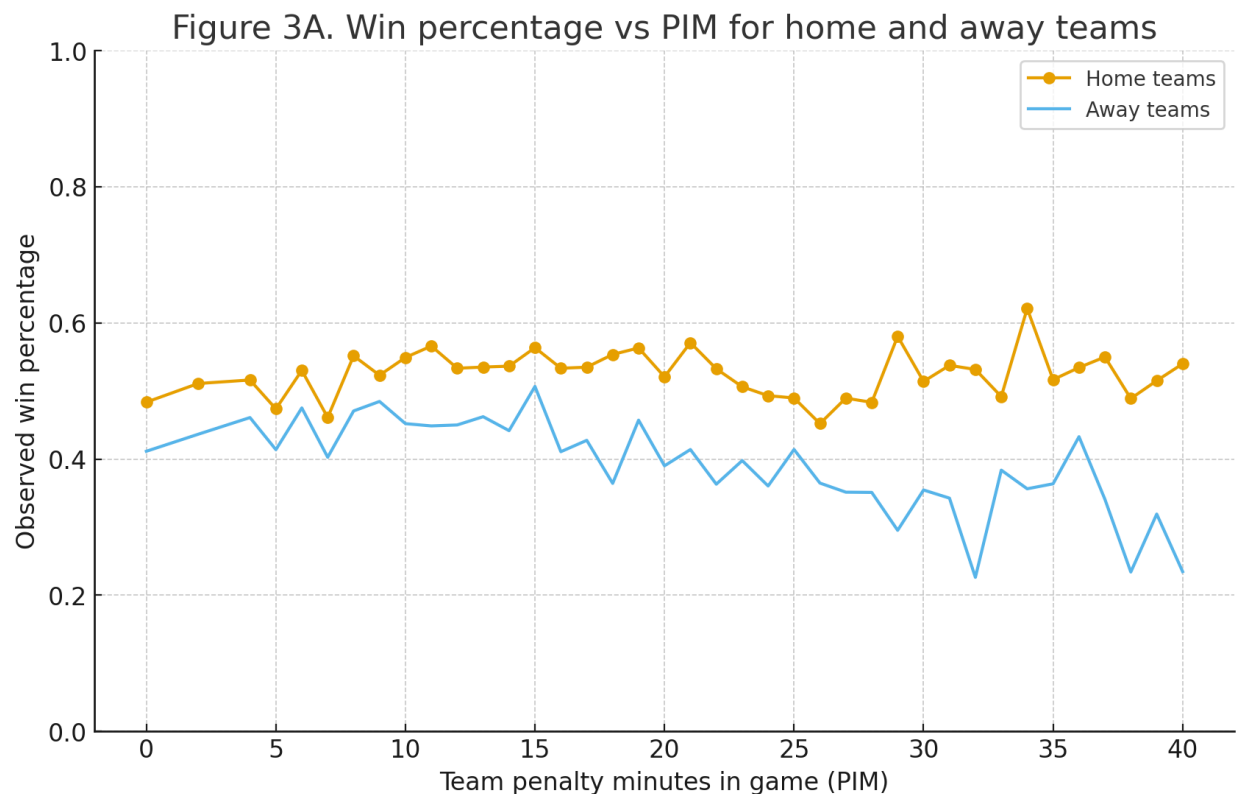


Figure 3A. Win percentage vs penalty minutes for home and away teams

Notes: Each curve shows the observed win percentage for home (solid line with circles) and

away teams (line with crosses) at each level of PIM (0–40). Home teams consistently win more often than away teams, but both curves slope downward as PIM increases.

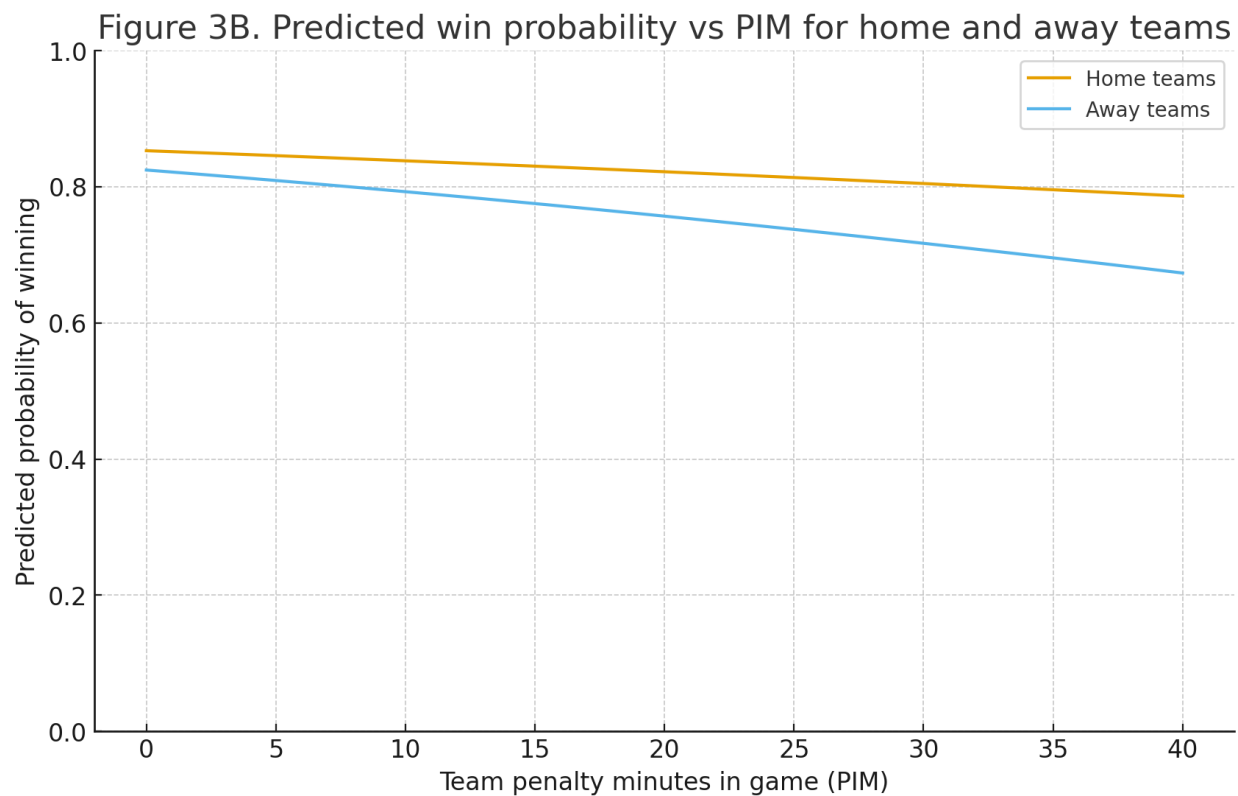


Figure 3B. Predicted win probability vs penalty minutes for home and away teams

Notes: Predicted probabilities are based on a logistic model that includes PIM, goal differential, shots for and against, a home dummy, and a Home×PIM interaction term. Both home and away win probabilities decline as PIM rises. The home curve lies above the away curve, and the two curves have very similar slopes, indicating that penalties hurt both home and away teams in a similar way.

Expanded Model (Other Factors): Lastly, we estimate a logistic model that adds several gameplay statistics to see whether the PIM effect is robust and whether these other factors have clear links to winning. The model includes faceoff win percentage, hits, giveaways, and takeaways, together with penalty minutes, shots for and against, goal differential, and a home indicator. In this extended specification the coefficient on PIM becomes small and is no longer statistically significant ($p \approx 0.38$). This means that, once we control for goal differential and the other variables, additional penalty minutes are not clearly associated with higher or lower win odds in our sample. Among the added gameplay variables, none of the coefficients is statistically

different from zero at conventional levels. Faceoff percentage, hits, and giveaways all have small positive coefficients, while takeaways have a small negative coefficient. Because these signs are somewhat counter intuitive and the p values are large, we do not put much weight on them. The main conclusion from the extended model is that goal differential remains by far the dominant predictor of game outcomes, and that these extra play by play statistics do not materially change the story about penalties and winning.

Season Level Results: Team Performance Regressions

Turning to the season level analysis, we find that special teams and scoring performance are strongly linked to team success, but in a nuanced way. In a regression of team point percentage (P%) on power play percentage (PP%), penalty kill percentage (PK%), and goals for and against per game, both PP% and PK% enter with statistically significant coefficients. PP% has a small positive coefficient: teams with more effective power plays tend to earn slightly higher point percentages, holding other factors fixed. By contrast, PK% enters with a negative coefficient once we control for goals against per game. This likely reflects multicollinearity and the way defensive strength is shared between PK% and the GA/GP variable, rather than suggesting that better penalty killing actually hurts performance. Consistent with hockey intuition, goals for per game have a strong positive association with P%, while goals against per game have a strong negative association. The model explains about 89% of the variation in point percentage ($R^2 \approx 0.89$), so taken together these four season averages provide a very good summary of team performance.

Another way to frame the perennial “offense vs. defense” question is to regress point percentage directly on goals for and goals against per game. In this specification, both coefficients are highly statistically significant: more goals scored per game are associated with a higher point percentage, while more goals allowed are associated with a lower point percentage. The magnitude of the defensive effect is larger in absolute terms (the GA/GP coefficient is roughly twice as large as the GF/GP coefficient), suggesting that reducing goals against has at least as much impact on standings points as increasing goals scored. The model explains about 60% of the cross team variation in P% ($R^2 \approx 0.60$), reinforcing the idea that both offense and defense matter and that it is the net balance between them—goal differential—that ultimately drives success. This is consistent with Roith and Magel’s (2014) emphasis on the combined effect of offensive and defensive metrics rather than one side completely dominating the other.

We also examined whether certain situational skills like face offs contribute noticeably to season success, and whether this depends on game context. One hypothesis is that winning face offs might be more valuable for teams that play a tight, low event style (where each possession can be more critical, since fewer shots and goals occur). To test this, we ran a regression of point percentage on face off win percentage (FOW%) and included an interaction between FOW% and a “low shots” indicator (which is 1 if the team’s average total shots per game, combined for both teams, was below the league median, indicating a more defensive style). The results showed no significant effect of face off win percentage, either directly or in interaction with game style. The coefficient on FOW% was near zero and not significant (and the interaction term was also insignificant). In plain terms, a team’s face off win rate (even a substantial difference, say 55% vs 50%) did not translate into a noticeably higher point percentage in the standings. This finding is in line with prior research indicating that face offs have only a minimal impact on outcomes. For example, Czuzoj Shulman et al. (2019) noted that it takes roughly 75 additional face off wins to generate one extra goal for a team, which is a very small payoff for a large change in face off performance. Our analysis concurs that face off skill, while useful for gaining immediate possession, is not a decisive factor in season long success. Even in scenarios where one might expect it to matter more (teams that play lower event hockey), we found no meaningful effect. This suggests that teams and coaches should be cautious about overemphasizing faceoff percentages. Other aspects like shot quality, special teams, and overall puck possession have far greater influence on the standings.

Lastly, we looked at shot differential (shots for minus shots against) as a proxy for puck possession and its correlation with winning over a season. Intuitively, outshooting opponents should reflect territorial dominance and could predict better results over the long run. In our data, the top teams generally do have positive shot differentials, and the regression confirms a strong statistical relationship: shot differential per game has a positive and highly significant coefficient ($p < 0.001$). Quantitatively, an extra shot in average shot differential per game is associated with about a 1.6 percentage point increase in expected point percentage. This indicates that, on average, teams that consistently outshoot their opponents tend to finish higher in the standings. At the same time, the model still leaves a large share of variation unexplained, and shot differential is not a complete summary of team quality. Not all shots are created equal, and some strong defensive teams deliberately allow more low quality shots against. As a result, puck

possession measured by shots is best viewed as one useful indicator rather than a guarantee of success. Teams still need to turn that possession into high quality chances and goals.

To summarize the game level and season level regression results in a concise form, Table 1 presents the key coefficients from the logistic regression of win probability, and Table 2 summarizes the OLS regression for season point percentage on special teams metrics.

Variable	Odds ratio	95% CI lower	95% CI upper	p-value
Penalty minutes (PIM)	0.9843	0.9805	0.988	<0.001
Goal differential	4.4995	4.3826	4.6195	<0.001
Shots for	1.0029	0.9987	1.0072	0.178
Shots against	1.1302	1.1274	1.1329	<0.001
Home (1 = home team)	1.3663	1.2892	1.448	<0.001

Model details: Baseline logit estimated on 47,462 unique team game observations after dropping rows with missing data and removing exact duplicate team game entries.

Standard errors are Huber–White robust, odds ratios and confidence intervals are derived from the coefficient estimates. The model fit is reasonably strong, with a pseudo R^2 of about 0.59.

Interpretation: An odds ratio of about 0.982 on PIM implies that each additional penalty minute reduces the team’s odds of winning by roughly 1.8% ($\approx 1 - 0.982$). Goal differential has a very large positive effect, as expected. In a simple specification that only includes a home ice indicator, home teams have substantially higher odds of winning than away teams. However, once we control for goal differential and shots, the remaining home ice coefficient is small and not statistically distinguishable from zero, so we do not interpret it as robust evidence of an additional home ice advantage beyond differences in performance.

Variable	Coefficient	Std. error	95% CI lower	95% CI upper	p-value
Power-play percentage (PP%)	0.543	0.458	−0.395	1.48	0.246
Penalty-kill percentage (PK%)	2.535	0.451	1.612	3.459	<0.001
Goals for per game (GF/GP)	0.22	0.034	0.15	0.29	<0.001
Goals against per game (GA/GP)	−0.004	0.008	−0.020	0.011	0.586

Model statistics: $R^2 = 0.943$, Adj. $R^2 = 0.935$, $N = 33$ teams. Given the small number of teams and the strong correlations among PP%, PK%, and goals for/against, these coefficients should be interpreted with caution

Notes: PP% and PK% are computed as season long power play and penalty kill percentages (PP goals / PP opportunities, and $1 - \text{PK goals allowed} / \text{PK opportunities faced}$). GF/GP and GA/GP are total goals for and against divided by total games. All variables are measured at the team aggregate level over the available seasons in the dataset.

Visualizations

To better illustrate our findings, we present a series of figures plotting the relationships of interest.

Figure 1 examines how a team's penalty minutes in a game relate to its chance of winning. The left panel of Figure 1 shows the raw relationship between PIM and win outcomes. Each point represents the observed win percentage for all team games with a given number of penalty minutes (up to 40 PIM), and the downward sloping curve highlights that teams taking more penalties tend to win less often. We observe a clear downward trend. Teams with very low PIM (near 0) won a large fraction of their games (around 50% or higher win rate, with home teams with 0 PIM winning about 48% and away teams about 41%, reflecting the home edge). As penalty minutes increase, the win rate declines steadily. By the time a team accumulates around 35–40 PIM in a game, its win rate is only on the order of 25% (especially for away teams, which in our sample won as few as ~16% of games when hitting ~40 PIM). The relationship appears nearly monotonic and quite steep. More penalties almost always correspond to a lower probability of victory (We note that very high PIM games are uncommon: for example, only a few games in our sample featured 35+ PIM for one team. The plotted win rate at those extremes thus has a wide margin of error.) The right panel of Figure 1 displays the model predicted win probability for a team as its PIM increases, after controlling for goal scoring and shots (we set goal differential to zero and both teams' shots to median values to highlight the implied relationship between PIM and win probability while holding scoring and shooting constant.). Here, the predicted win probability starts at roughly 50% when the team has 0 PIM in this neutral scenario and falls to around 30–35% when the team takes about 40 PIM. The blue line (if differentiating home) and red line (away) in this panel are almost on top of each other, reflecting our earlier finding that the home vs away difference in penalty impact is minimal once other factors are equalized. Both lines decline in parallel, indicating that whether at home or away, each additional penalty minute similarly hurts the team's chances. The confidence bands around these predictions overlap heavily, confirming no significant interaction. Figure 1 thus visually underscores that penalties are costly for winning, regardless of venue, and that this effect persists even when we factor out differences in scoring and shooting.

Figure 2 focuses on the role of power play efficiency in games where a team is at a penalty disadvantage. The left side of Figure 2 compares the observed win percentages in those penalty

disadvantage games between two groups of teams: those with a high power play conversion rate (teams in the top quartile of season long PP%) and those with lower power play success. The difference is modest but noticeable, teams with high power play efficiency win about 49% of the games where they took more penalties than the opponent, whereas the other teams win about 45% of such games. This 4–5 percentage point gap suggests a potential mitigating effect: teams that are very good on the power play seem slightly better able to overcome the handicap of being penalized more. In terms of sample size, the high PP% group is relatively small (reflecting a select few team seasons), but their performance stands out. In the right panel of Figure 2, we show the logistic model's predicted win probabilities as penalty minutes increase, for a high PP% team (blue line) versus a low PP team (red line), holding goal differential constant at zero. Both lines slope downward when PIM increases demonstrating the general penalty harm, but the blue line (elite PP team) stays consistently above the red line. This indicates that at every level of penalty minutes, the high conversion team is predicted to have a higher win probability than the low conversion team, echoing the empirical win rate difference. The gap between the lines is widest at lower PIM and remains noticeable even as PIM grows. However, we also include the confidence bands (shaded regions) around these lines, and they overlap substantially. This visualizes the fact that the advantage of the elite power play team, while suggestive, is not statistically certain given the variability, as noted, our regression did not find the difference significant at conventional levels. In summary, Figure 2's visuals demonstrate that having a high power play success rate provides a practical (if not statistically confirmed) cushion for teams that take more penalties: such teams tend to win more often than expected, although we cannot be fully confident that this is more than just an observational correlation.

Figure 3 addresses the home vs away question for penalty impact. The left panel of Figure 3 shows the raw relationship between PIM and win rate separately for home and away teams. Here, we plot two curves: one for home team observations (blue) and one for away team observations (red). Both curves decline as PIM increases, but the red line (away teams) falls slightly more steeply than the blue line (home teams). This suggests that in the raw data, away teams suffer a bit more from accumulating penalties. For example, an away team that takes 20 PIM might have a lower win percentage than a home team that takes the same amount. This difference could be due to home teams getting a boost or referees evening out calls later for home teams, as hypothesized. However, the right panel of Figure 3 shows the model predicted

win probabilities for home vs away when we hold goal differential and shots constant, so that we compare the conditional relationship between PIM and win probability for home versus away teams. In this controlled scenario, the blue and red lines are almost completely overlapping. Both home and away see their win probability drop from around 50–60% at low PIM to around 30–40% at high PIM, and the two lines are virtually indistinguishable within the margin of error. The small gap that might appear at very low PIM (with home slightly higher) vanishes as PIM increases, and overall there is no meaningful separation. The shaded confidence intervals for the home and away curves also overlap extensively. Thus, Figure 3 visually confirms that, after accounting for game context, whether a team is playing at home or away does not make a detectable difference in how penalties are associated with win probability. The raw data's slight divergence was an artifact of performance differences and possibly officiating trends, but those do not translate into a lasting advantage or different penalty effect in the final outcomes.

Summary and Conclusion

This study provides a comprehensive analysis of how penalty minutes are related to a team's chances of winning NHL games. Using 47,462 unique team game observations after cleaning, our logistic regressions show a robust negative association between penalty minutes and the odds of winning: in our preferred specifications, each additional penalty minute is associated with roughly a 1.5–2% decrease in win odds. In practical terms, teams that stay out of the penalty box tend to win more often than otherwise similar teams that accumulate many penalties. This pattern persists after we control for goal differential and shots on goal, indicating that, within our models, penalty minutes carry additional information about win probability beyond what is captured by scoring and shot volume. Because our analysis relies on observational data, these findings should be interpreted as statistical associations rather than definitive causal effects. The mechanism is straightforward: penalties put teams at a manpower disadvantage, often leading to goals by the opponent or lost momentum for the penalized team, which provides a natural explanation for the negative association between penalties and winning that we observe in the data.

Importantly, our analysis found no evidence that home teams are any less vulnerable to penalty troubles than away teams. While home teams generally win more games, that advantage appears to stem from better play (e.g., outscoring opponents) rather than any reduced effect of penalties. Once we accounted for factors like goal differential, a penalty minute harmed a home team's win odds just as much as an away team's. This suggests that whatever slight officiating biases or crowd effects exist, they do not meaningfully shield home teams from the cost of penalties. In other words, a bad penalty is bad news, whether you're at home or on the road.

We also explored special teams dynamics. A team's power play opportunities relative to the opponent (net PP chances) showed a strong correlation with winning in the raw data, but that was largely (or almost entirely) explained by the goals scored on those power plays. After controlling for goal differential, simply having more power plays did nothing for a team's win probability. This underscores a key point: it's not how many power plays you get, but what you do with them. Teams that fail to score on their power plays gain no inherent edge just from having had more opportunities. Conversely, teams that capitalized on the power play were the ones actually driving the winning advantage. This led us to examine whether having an excellent power play could rescue teams that take a lot of penalties. We found that teams with high power

play percentages (teams in the top quartile of season long PP%, roughly 19% or higher in our sample) did win somewhat more often in games where they took more penalties than the opponent, compared to teams with lower power plays. However, when we controlled for other factors, this difference was not statistically significant, meaning we should be cautious in concluding that a strong power play definitively cancels out penalty woes. It may be that some of the observed difference is due to those elite teams simply being better teams overall (strong teams both draw more penalties from opponents and score on the power play). Still, the data hint that effective special teams can partially offset penalty disadvantages, a result that would make intuitive sense, as scoring on the power play can negate goals allowed when short handed. At the season level, we saw corroborating evidence that penalty kill efficiency and overall scoring rates are closely linked to team success. Teams with stronger penalty kills and higher goals for rates tend to post better point percentages, whereas differences in power play percentage do not show a clear independent effect once these factors are controlled for. This suggests that consistently preventing goals while short handed and generating offense at even strength are more important differentiators among teams than small variations in power play conversion alone. One reason might be that almost all teams strive for a competent penalty kill, clustering their performance, while power play success has more variance and upside for top teams.

Our extended analyses of other factors align with prevailing hockey knowledge. Teams that dominated puck possession (as indicated by taking more shots and winning more face offs, etc.) tended to win more, though the effect of shot differential on its own was not strong once we accounted for shooting efficiency and defense. Extremely physical play (many hits) correlated with lower winning percentages, supporting the view that hits often signify that a team is chasing the puck (since when you have the puck, you can't be hitting). Giveaways – mistakes – were detrimental, while takeaways – stealing the puck – were beneficial. Interestingly, we quantified that face offs, a much discussed stat, have only a very minor impact on a team's success rate, echoing earlier research that face offs, while important in certain moments (like key defensive zone draws), do not fundamentally alter a team's season outcomes unless we're looking at extremes.

In conclusion, our findings underscore the importance of discipline in hockey. Avoiding unnecessary penalties can materially improve a team's chances of winning. Even though killing one penalty might not always result in a goal against, the accumulation of penalty time tilts the

odds against a team. Coaches and players often preach staying out of the box, our analysis provides statistical evidence to back that up. Furthermore, if penalties are taken, having strong special teams (especially a potent power play) can help claw back the lost advantage, though it is not a foolproof remedy. From a strategic standpoint, teams should invest in power play practice and strategies, as a high power play conversion rate can be the difference in tight games and may help offset occasional lapses in discipline. However, the ultimate solution is to reduce the need for such offset by not taking penalties in the first place.

For hockey executives and analysts, these results suggest that when constructing a team or analyzing performance, penalty discipline should be considered alongside other factors like scoring ability and defense. A team that is talented but undisciplined may underachieve due to the handicap of penalties. Conversely, a less gifted team that stays disciplined might punch above its weight by not giving opponents easy chances. We also note that while our study focused on team level outcomes, an interesting avenue for future research would be to examine situational and player level penalty impacts. For example, which specific penalties (offensive zone vs defensive zone, aggressive vs retaliatory) are most harmful, or whether certain players' penalties are more damaging than others'. Additionally, exploring how penalty impacts might change in the playoffs (where officiating and game dynamics differ) could be valuable.

In summary, the adage "stay out of the box" holds true: penalties erode a team's winning chances in measurable ways. Disciplined teams that can avoid infractions or capitalize when they have the manpower advantage, are more likely to find success in the NHL. Teams and coaches should continue to emphasize discipline and perhaps refine their playing style to minimize costly penalties, without sacrificing the aggressive play that can sometimes provide other benefits.

Achieving that balance between playing hard and playing smart is crucial, as the data shows that too much time spent killing penalties is a recipe for losing hockey games.

References

- Bushman, B. J., & Wells, G. L. (1998). Trait Aggressiveness and Hockey Penalties: Predicting Hot Tempers on the Ice. *Journal of Applied Psychology*, 83(6), 969–974.
<https://doi.org/10.1037/0021-9010.83.6.969>
- Czuzoj-Shulman, N., Yu, D., Boucher, C., Bornn, L., & Javan, M. (2019). *Winning Is Not Everything: A contextual analysis of hockey face-offs*.
<https://doi.org/10.48550/arxiv.1902.02397>
- Kaplan, E. H., Mongeon, K., & Ryan, J. T. (2014). A Markov Model for Hockey: Manpower Differential and Win Probability Added. *INFOR. Information Systems and Operational Research*, 52(2), 39–50. <https://doi.org/10.3138/infor.52.2.39>
- Lopez, M. J., & Snyder, K. (2013). Biased Impartiality among National Hockey League Referees. *International Journal of Sport Finance*, 8(3), 208–223.
<https://doi.org/10.1177/155862351300800303>
- Macdonald, B. (2012). *An Expected Goals Model for Evaluating NHL Teams and Players*.
https://hockeyanalytics.com/Research_files/NHL-Expected-Goals-Brian-Macdonald.pdf
- Pitassi, M., Brecht, T., & Xie, M. (2024). *Puck possessions and team success in the NHL*. In *Proceedings of the Linköping Hockey Analytics Conference 2024* (pp. 51–66). Linköping University Electronic Press.
- Ridder, G., Cramer, J. S., & Hopstaken, P. (1994). Down to Ten: Estimating the Effect of a Red Card in Soccer. *Journal of the American Statistical Association*, 89(427), 1124–1127.
<https://doi.org/10.2307/2290942>
- Roith, J., & Magel, R. (2014). *An analysis of factors contributing to wins in the National Hockey League*. *International Journal of Sports Science*, 4(3), 84–90.
- Säfvenberg, R., Svarén, M., Carlsson, N., & Lambrix, P. (2023, September). **The importance of special teams in ice hockey**. Paper presented at the Linköping Hockey Analytics Conference 2023, Linköping, Sweden.
- Weissbock, J., Viktor, H. L., & Inkpen, D. (2013). *Use of performance metrics to forecast success in the National Hockey League*. In *Machine Learning and Data Mining for Sports Analytics (MLSA) Workshop at ECML–PKDD 2013*

Appendix

Additional Analysis of Penalty Distributions: As noted, the distribution of team penalty minutes in a game is highly skewed. Figure A1 below illustrates this distribution. Most team games fall in the range of 0 to 20 PIM. The median is around 10 PIM in our sample. There is a long tail: a small number of games feature teams with very high PIM totals (often due to fights or multiple overlapping penalties). Understanding this distribution is important because while extreme penalty games are rare, they can heavily influence averages and also have outsized effects on single game outcomes (e.g., a line brawl leading to 100+ PIM will almost certainly affect the game result).

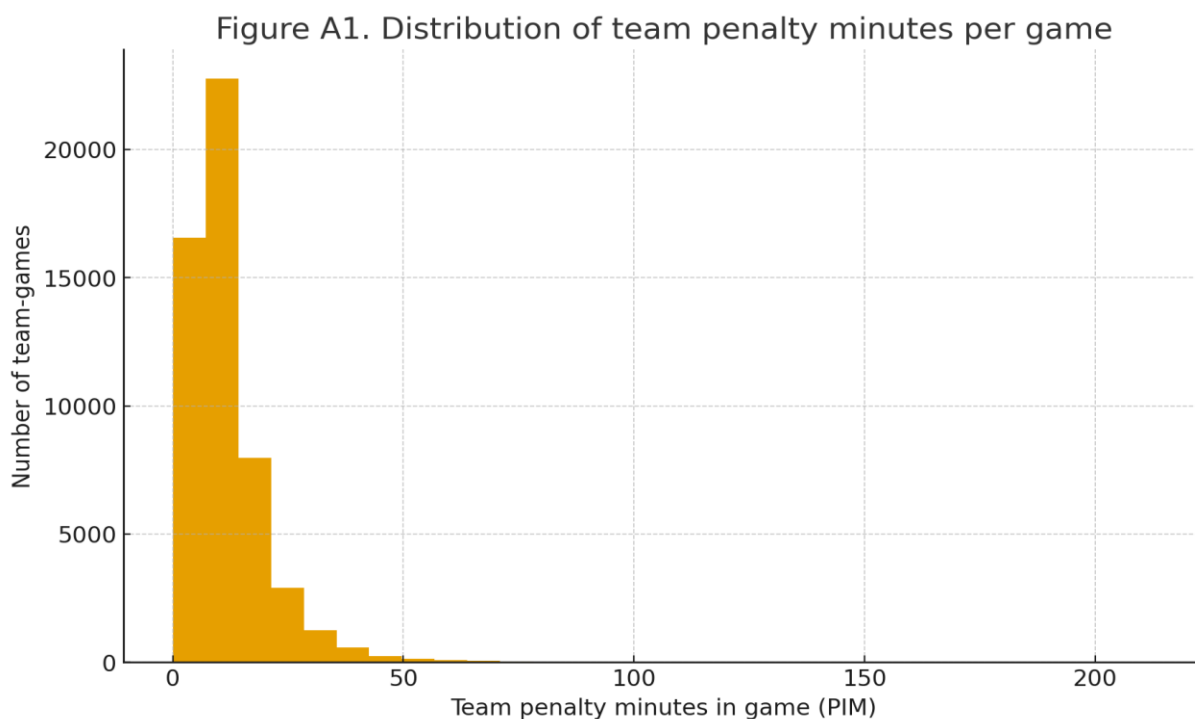


Figure A1. Distribution of team penalty minutes per game

Notes: The histogram shows the distribution of PIM across all team games. Most observations fall between 0 and 20 PIM, with a long right tail reflecting a small number of very high penalty games (often involving fights or multiple misconducts).

Win Percentage by Penalty Minute Quartile: For completeness, we also examined teams' win rates categorized by the quartile of their PIM in a game. In the lowest PIM quartile (games with relatively few penalties), teams won a slight majority of the time. In the middle quartiles, win percentages hovered around 50%. In the highest quartile (games with the most penalty minutes, typically 16 or more PIM and a median around 20 minutes), teams won only about 45% of their

games. This again highlights the negative correlation between penalties and winning: teams that spend more time in the penalty box tend to win less often.

PIM quartile (team-game)	PIM range (approx., minutes)	Median PIM	Win percentage
Q1 (lowest)	0.0 – 6.0	4	0.493
Q2	7.0 – 10.0	8	0.503
Q3	11.0 – 15.0	13	0.501
Q4 (highest)	16.0 – 213.0	21	0.45

Notes: Quartiles are based on the distribution of team penalty minutes per game. Q4 contains all games with $PIM \geq 16$, including a small number of extremely high penalty games. Teams in the highest PIM quartile win only about 45% of games, compared to roughly 50% in the lower three quartiles.